

JavaScript Objects & Higher Order Functions - Lab Assignment

Topics Covered:

- Objects
- Array of Objects
- Functions
- map()
- filter()
- reduce()
- sort()
- forEach()

Instructions:

1. Use Higher Order Functions wherever possible.
2. Create separate functions for each task.
3. Write clean and readable code.
4. Display output properly.

Lab 1: Student Result Management System

```
const students = [  
  {id:1,name:'Rahul',marks:85,course:'MERN'},  
  {id:2,name:'Priya',marks:92,course:'Java'},  
  {id:3,name:'Aman',marks:65,course:'Python'},  
  {id:4,name:'Neha',marks:78,course:'MERN'},  
  {id:5,name:'Rohit',marks:95,course:'Java'}  
]
```

Tasks:

1. Display all student names using map().
2. Find students scoring above 80 using filter().
3. Calculate total and average marks using reduce().
4. Find topper.
5. Sort students by marks descending.
6. Count students course-wise.

Expected Output:

Topper: Rohit (95)

Average Marks: 83

Lab 2: E-Commerce Product Dashboard

```
const products = [  
  {id:1,name:'Laptop',price:70000,stock:10},  
  {id:2,name:'Phone',price:25000,stock:15},  
  {id:3,name:'Headphone',price:3000,stock:20},  
  {id:4,name:'Watch',price:5000,stock:12}  
]
```

Tasks:

1. Display all product names.
2. Find products costing above 10000.
3. Calculate inventory value (price × stock).
4. Find most expensive product.
5. Sort by price descending.
6. Create a new discounted product list using map().

Expected Output:

Most Expensive Product: Laptop

Lab 3: Employee Management System

```
const employees = [  
  {id:101,name:'Amit',salary:35000,department:'HR'},  
  {id:102,name:'Riya',salary:50000,department:'IT'},  
  {id:103,name:'Karan',salary:45000,department:'IT'},  
  {id:104,name:'Pooja',salary:60000,department:'Finance'}  
]
```

Tasks:

1. Display employee names.
2. Find employees earning above 45000.
3. Find highest paid employee.
4. Calculate total salary expense.
5. Count employees department-wise.
6. Sort employees by salary.

Expected Output:

Highest Paid Employee: Pooja

Lab 4: Online Order Processing System

```
const orders = [  
  {orderId:1,customer:'Rahul',amount:1200},  
  {orderId:2,customer:'Priya',amount:800},  
  {orderId:3,customer:'Aman',amount:2500},  
  {orderId:4,customer:'Neha',amount:1500}  
]
```

Tasks:

1. Display all customer names.
2. Find orders above Rs.1000.
3. Calculate total revenue.
4. Calculate average order value.
5. Apply 5% discount using map().
6. Sort orders by amount.

Expected Output:

Total Revenue: 6000

Average Order Value: 1500

Lab 5: University Management System (Comprehensive Revision)

```
const students = [  
  {id:1,name:'Rahul',course:'MERN',fees:45000,attendance:85},  
  {id:2,name:'Priya',course:'Java',fees:50000,attendance:92},  
  {id:3,name:'Aman',course:'Python',fees:35000,attendance:70},  
  {id:4,name:'Neha',course:'MERN',fees:45000,attendance:95},  
  {id:5,name:'Rohit',course:'Java',fees:50000,attendance:88}  
]
```

Tasks:

1. Display all student names.
2. Find students with attendance above 80.
3. Calculate total fees collected.
4. Calculate average attendance.
5. Find highest attendance student.
6. Sort students by attendance.
7. Count students course-wise.
8. Create a new array containing only names.
9. Search student by name.
10. Generate complete report.

Expected Output:

Highest Attendance: Neha (95)

Total Fees Collected: 225000

MERN: 2

Java: 2

Python: 1